SE LAB 7

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Section: 5H

Buggy module run (python credit_card_rewards_parser.py)

Valid card number and transaction are handled, but entering the statement total without “Total:” makes parse_statement_total crash with an IndexError.

```
(.venv) PS C:\00 College work\5th SEM\SE\LAB 7\L6_L7_61_credit_card_rewards\LAB7> python credit_card_rewards_parser.py
==== Running mask_card_number ====
Enter a card number (e.g. 4111 1111 1111 1234): 4111 1111 1111 1234
Masked card number: ************1234

==== Running parse_transaction ====
Enter a transaction as DATE,CATEGORY,AMOUNT (e.g. 2026-11-05,dining,1550): 2026-11-07,dining,400
Category and amount: ('dining', 400)

==== Running parse_statement_total ====
Enter the statement total line (e.g. Total: 12,500): 12500
Traceback (most recent call last):
  File "C:\00 College work\5th SEM\SE\LAB 7\L6_L7_61_credit_card_rewards\LAB7\credit_card_rewards_parser.py", line 39, in <module>
    main()
  File "C:\00 College work\5th SEM\SE\LAB 7\L6_L7_61_credit_card_rewards\LAB7\credit_card_rewards_parser.py", line 36, in main
    print("Statement total:", parse_statement_total(value))
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\00 College work\5th SEM\SE\LAB 7\L6_L7_61_credit_card_rewards\LAB7\credit_card_rewards_parser.py", line 22, in parse_statement_total
    amount = text.split(':')[1]
             ~~~~~~~~~~~~~~~^^^
IndexError: list index out of range
(.venv) PS C:\00 College work\5th SEM\SE\LAB 7\L6_L7_61_credit_card_rewards\LAB7>
```

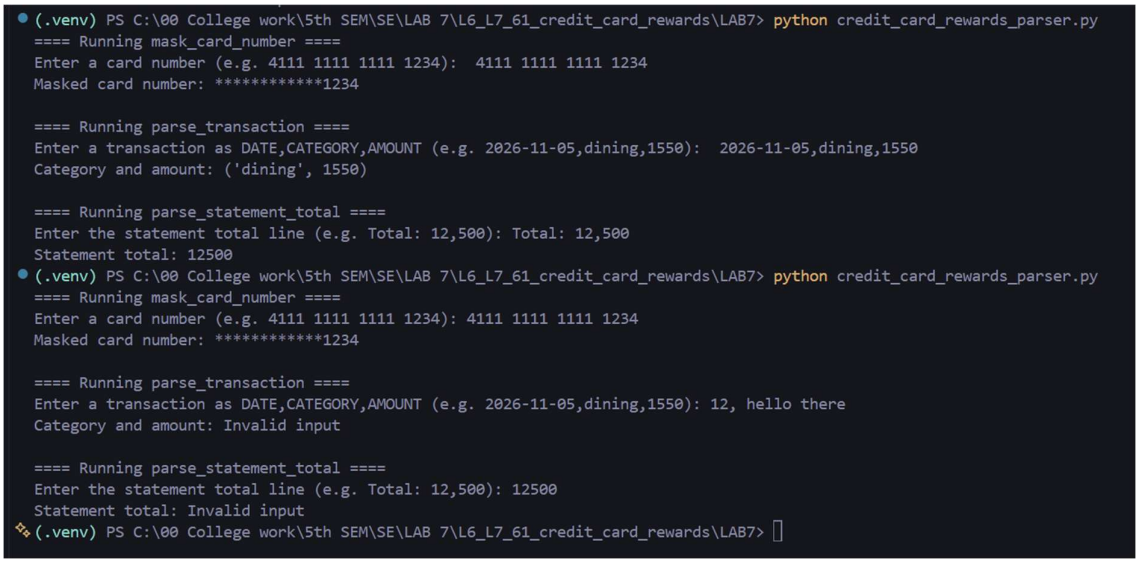
Hypothesis tests on the buggy module (pytest test_credit_card_rewards_parser.py)

All 3 tests fail. mask_card_number crashes on None (AttributeError); Hypothesis found 2 distinct failures in parse_transaction and 3 in parse_statement_total.

```
=========================== short test summary info ===========================
FAILED test_credit_card_rewards_parser.py::test_mask_card_number_no_crash - AssertionError: mask_card_number crashed with: AttributeError("'NoneType' object has no attribute 'replace'")
FAILED test_credit_card_rewards_parser.py::test_parse_transaction_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_credit_card_rewards_parser.py::test_parse_statement_total_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================== 3 failed in 0.35s ==============================
(.venv) PS C:\00 College work\5th SEM\SE\LAB 7\L6_L7_61_credit_card_rewards\LAB7>
```

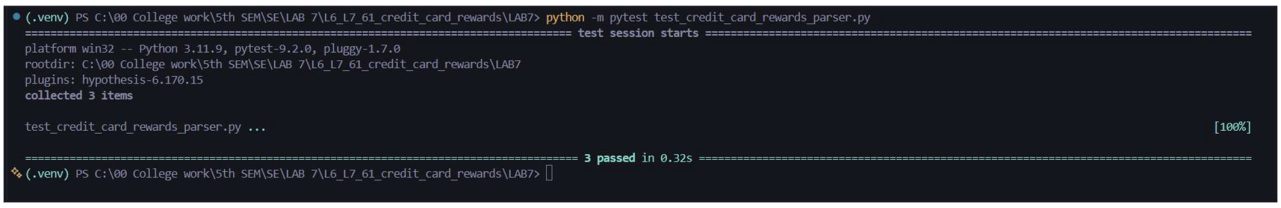
SS3: Fixed module run (python credit_card_rewards_parser.py)

Valid inputs still give the right results. Invalid inputs (a transaction with missing fields, a total with no colon) now print “Invalid input” instead of crashing.



SS4: Hypothesis tests on the fixed module (pytest test_credit_card_rewards_parser.py)

All 3 tests pass.



Bugs found and fixes

- **mask_card_number:** crashed on None and accepted junk such as “ab”. Now returns None unless the input is a string of 12 to 19 digits (spaces and dashes allowed).
- **parse_transaction:** crashed on None (AttributeError), on lines with fewer than three fields (IndexError) and on non-numeric amounts (ValueError). Now checks for exactly three fields, a non-empty category and a whole-number amount.
- **parse_statement_total:** crashed on None, on lines with no colon (IndexError) and on non-numeric amounts (ValueError). Now checks for the colon and a valid number after it.

- **Both parsers:** int() also accepts non-ASCII digits (e.g. Arabic-Indic digits) and fails on strings over 4300 digits, so amounts are matched against plain 0-9 digits and the ValueError is caught.